Continuous Assessment 3 (CA-3) - Practice Questions

Course Code: INT396 | Course Title: UNSUPERVISED LEARNING

Section(s): K3P24MK, K3P24WV | Syllabus Covered: Unit 3 and Unit 4

CA-3 pattern: 3 questions x 10 marks = 30 | Q1 Unit 3, Q2 Unit 4, Q3 half Unit 3 and half Unit 4

Instructions: prepare every question below. Each carries 10 marks; the marks for each part are shown in brackets. Write structured answers, show all working for numerical parts, and draw diagrams wherever they help.

Unit 3: Hierarchical and Density-Based Clustering

1. A city council clusters 40 neighbourhoods by their facilities using agglomerative hierarchical clustering and studies the resulting dendrogram. (10 Marks)

(a) Explain what a dendrogram shows and what the height of a merge means. [5]

(b) Explain how a cut is chosen to decide the number of clusters, and state one limitation of hierarchical clustering on a very large dataset. [5]

2. DBSCAN is run on the one-dimensional points 1, 2, 3, 10, 11, 12 and 25, with eps = 1.5 and MinPts = 3 (counting the point itself). (10 Marks)

(a) Identify the core points, showing the neighbours of each point. [5]

(b) Identify the clusters that are formed and any noise points, and state what would happen to the point 25 if eps were raised to 15. [5]

3. Five shops lie on a straight road at positions 1, 2, 6, 7 and 15 km and are to be grouped by agglomerative hierarchical clustering. (10 Marks)

(a) Explain the steps that agglomerative hierarchical clustering follows, from the start until only one cluster is left. [5]

(b) Using single linkage, list the merges in the order they happen, with the distance at which each merge takes place, and state how many clusters are obtained, with their members, if the dendrogram is cut at a height of 3. [5]

4. Agglomerative hierarchical clustering needs a linkage criterion to measure the distance between two clusters. For two particular clusters, the distances between their members are 3, 5, 8 and 10. (10 Marks)

(a) Describe single linkage, complete linkage and average linkage. [5]

(b) Find the single-linkage, complete-linkage and average-linkage distances between the two clusters. [5]

5. A logistics firm groups delivery GPS readings with DBSCAN, which forms clusters from dense regions of points. (10 Marks)

(a) State what the two parameters eps and MinPts control, and define a core point, a border point and a noise point. [5]

(b) Draw a simple sketch showing one point of each kind, and give three advantages of DBSCAN over k-Means. [5]

Unit 4: Dimensionality Reduction and Representation Learning

6. A hospital dataset describes each patient by 900 features, many of them measured in different units. (10 Marks)

(a) Explain what is meant by the curse of dimensionality, and why features measured in different units must be standardised before Principal Component Analysis is applied. [5]

(b) State the difference between Principal Component Analysis and manifold learning when the data lies on a curved surface, and give one use of manifold learning. [5]

7. An analyst must plot a dataset of 50 features in two dimensions and is choosing between Principal Component Analysis and t-SNE. (10 Marks)

(a) State what Principal Component Analysis preserves and what t-SNE preserves. [5]

(b) State one situation in which each of the two is the better choice, with a reason. [5]

8. A company records many correlated features for each customer and wants to reduce them with Principal Component Analysis. (10 Marks)

(a) State what dimensionality reduction is, and give two reasons why it is needed. [5]

(b) Describe the main steps of Principal Component Analysis. [5]

9. An autoencoder is trained to compress images of handwritten digits into a short code and then rebuild them. (10 Marks)

(a) Draw and label the encoder, the bottleneck and the decoder. [5]

(b) Explain what the network is trained to minimise, and give one difference between an autoencoder and Principal Component Analysis. [5]

10. An analyst applies Principal Component Analysis to a dataset of 60 features and must decide how many components to keep. (10 Marks)

(a) Explain what the explained variance of a principal component tells you, and how it is used to decide how many components to keep. [5]

(b) Explain what a scree plot shows and how it is read. [5]

Units 3 and 4: Combined Questions (part (a) from Unit 3, part (b) from Unit 4)

11. A mobile company records 40 usage features for each of 20,000 customers. It wants to find groups of similar customers, set aside the few unusual ones, and show the groups on a two-dimensional plot. (10 Marks)

(a) Compare k-Means, agglomerative hierarchical clustering and DBSCAN on whether the number of clusters must be fixed in advance, and state which of the three suits this data, with the reason. [5]

(b) State what UMAP is used for, and give two differences between UMAP and t-SNE. [5]

12. A delivery firm wants to find busy pick-up zones with DBSCAN. Besides the location of each pick-up it has recorded several other features, and it is unsure how many of them to keep. (10 Marks)

(a) Explain how a k-distance plot is drawn and how it is used to choose eps for DBSCAN. [5]

(b) Explain the curse of dimensionality, and how it affects distance-based clustering such as DBSCAN. [5]

13. An analyst groups customers with agglomerative hierarchical clustering. Two of the customer features, monthly spend and number of orders, rise and fall together, so the features are first reduced with Principal Component Analysis. (10 Marks)

(a) Explain how Ward linkage measures the distance between two clusters, and state why complete, average or Ward linkage is often preferred to single linkage. [5]

(b) Explain why Principal Component Analysis is useful when two features rise and fall together. [5]

14. Five weather stations stand along a highway at 1, 2, 4, 8 and 9 km. Each station records 50 readings every hour, and the readings are to be compressed with an autoencoder. (10 Marks)

(a) Using complete linkage on the positions of the stations, list the merges in the order they happen, with the distance at which each merge takes place. [5]

(b) Explain the role of the bottleneck in the autoencoder, and what happens if the bottleneck is made too narrow or too wide. [5]

15. A research team plans to group patients with DBSCAN. Some patient groups are tightly packed and others are spread out, and each patient is described by four standardised measurements that are first reduced with Principal Component Analysis. (10 Marks)

(a) State two situations in which DBSCAN does not work well, with the reason for each. [5]

(b) Explain what the first principal component represents, and why the measurements are standardised before Principal Component Analysis. [5]

----- End of Practice Questions -----